

Database Schema & Implementation Roadmap

Enterprise-Grade Workflow Analytics Tracker Platform for BPO Operations

Author: Product Architecture Team

Target Audience: Engineering Lead, Database Architects, COO

1. Data Architecture Strategy

To accurately scale an operational analytics framework to thousands of remote agents, a single standard database will fail. High-frequency agent tracking data creates intensive write strain, while analytics engines require ultra-fast reads. We implement a **polyglot storage strategy**:

- **PostgreSQL (Relational Core):** System configuration, tenancy rules, access logs, and structural business relationships.
- **ClickHouse (Time-Series Analytics Engine):** High-throughput append-only data store designed for processing millions of metrics (keystroke counts, click rates, app focused events) directly from the desktop agent.

2. Relational Schema (PostgreSQL Core)

The relational model defines operational structures such as multi-tenancy, workforce hierarchy, and baseline productivity rules definitions.

Table: users

Column Name	Type	Constraints / Notes
id	UUID	PRIMARY KEY, generated default uuid_generate_v4()
tenant_id	UUID	FOREIGN KEY -> tenants(id), indexing for scoping partitioned data
email	VARCHAR(255)	UNIQUE, operational login, filtered lookup optimization
password_hash	VARCHAR(255)	Argon2id securely salted hash storage secure element
role	VARCHAR(32)	ENUM: 'agent', 'team_leader', 'ops_manager', 'coo', 'admin'
manager_id	UUID	NULLABLE FOREIGN KEY -> users(id) for organization nesting hierarchy
created_at	TIMESTAMPTZ	Default clock now system anchor timestamp

Table: application_registry

Column Name	Type	Constraints / Notes
id	BIGSERIAL	PRIMARY KEY
tenant_id	UUID	FOREIGN KEY -> tenants(id)
process_name	VARCHAR(255)	Exact OS process image name (e.g., 'chrome.exe', 'crm_prod.exe')
window_title_regex	VARCHAR(512)	NULLABLE, regex matching string to extract deep target pathing
classification	VARCHAR(32)	ENUM: 'productive', 'non_productive', 'neutral'

3. Analytical Schema (ClickHouse Time-Series)

Optimized for continuous multi-row appending stream tracking events sent via the desktop agent every 10-30 seconds.

Table: agent_activity_logs

Field Name	Engine Type	Functional Explanation
timestamp	DateTime64(3)	UTC millisecond precision timestamp event resolution anchor
tenant_id	UUID	Primary structural grouping identity partitioning
agent_id	UUID	Unique tracking target user identifier match
process_name	LowCardinality(String)	Optimized string compression dictionary mapping process string
window_title	String	Active title header textual snapshot captured from context focus
keystroke_count	UInt16	Aggregated raw count of keystrokes across current loop interval
mouse_click_count	UInt16	Aggregated pointer click interactions captured
mouse_movement_px	UInt32	Summation delta tracker vector scalar measuring distance active
is_idle	UInt8	Boolean flag (0/1) indicating timeout thresholds broken

Engine Declaration Architecture Configuration:

```
ENGINE = MergeTree()  
PARTITION BY toYYYYMM(timestamp)  
ORDER BY (tenant_id, agent_id, timestamp)  
SETTINGS index_granularity = 8192;
```

4. Implementation Roadmap & Milestones

A 4-phase technical roadmap targeting production launch within 6 months.

Phase & Timeframe	Key Technical Deliverables & Milestones	Risk Mitigations
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Phase 1: Core Agent v1 Month 1 - 2	• Native OS Hook system (Windows/C#) tracking process focus. • Secure encrypted local queue store SQLite loop backup cache layer. • Base backend Ingestion API endpoint handling data package bursts.	Obtain EV Code Signing certificates early to stop smart antivirus blocking deployments.
Phase 2: Data Pipeline Month 3	• Deploy scalable ClickHouse data clustering backend nodes. • Build analytical data processing jobs converting raw streams into interval summaries. • Create real-time engine processing logic classifying process titles.	Establish early load stress-testing rigs simulating 5,000 active concurrent agent payloads.
Phase 3: Web Portal Month 4 - 5	• React dashboard interface construction for TL and Operations views. • Time-slice metrics rendering charts mapping out team timeline lanes. • System tracking manager hierarchy nesting tools implementation.	Enforce server cache indexes strictly on time filters to keep metrics pages responsive.
Phase 4: Launch Month 6	• Live controlled Beta trials launch tracking 200 production pilot users. • Stress validation security auditing penetration threat analysis execution. • Operational rollout platform support configuration baseline deployment.	Configure gradual client update groups rollout to prevent systemic downtime peaks.